

Detecting Hallucination in Bangla Large Language Model Responses: A Comparative Study of Bag-of-Words and TF-IDF Representations with Classical Machine Learning Classifiers

Md. Asraf Uddin

Department of Computer Science
American International University-
Bangladesh
Dhaka, Bangladesh

Farhan Chowdhury

Department of Computer Science
American International University-
Bangladesh
Dhaka, Bangladesh

Md. Shihab Uddin

Department of Computer Science
American International University-
Bangladesh
Dhaka, Bangladesh

Md. Shahriyar Hasan

Department of Computer Science
American International University-
Bangladesh
Dhaka, Bangladesh

Tohedul Islam

Department of Computer Science
American International University-
Bangladesh
Dhaka, Bangladesh

Abstract—Large language models (LLMs) are increasingly used for question answering, yet they frequently produce hallucinated responses: text that is fluent and plausible but unsupported by, or contradictory to, a given reference. This risk is amplified for low-resource languages such as Bangla, where dedicated hallucination-detection resources remain scarce. This paper presents a lightweight, classical machine-learning approach to detecting hallucination in Bangla question answering. We construct a labeled dataset of Bangla question, reference-answer, and LLM-answer triples, each annotated as Faithful or Hallucinated. The reference and LLM answers are concatenated into a single text unit, cleaned with Bangla-specific normalization rules, and tokenized into unigram and bigram features. Two lexical representations, Bag-of-Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF), are compared across three classifiers: Multinomial Naive Bayes, L1-regularized (LASSO) Logistic Regression, and a linear-kernel Support Vector Machine (SVM). To prevent data leakage from repeated questions, the dataset is split at the question level rather than the row level. Experimental results show that the TF-IDF + SVM configuration performs best, achieving an F1-score of 0.8016 and an accuracy of 73.58% on the Hallucinated class, and that TF-IDF outperforms BoW on average across all three classifiers (mean F1 of 0.7324 versus 0.7249). These findings indicate that simple lexical-overlap features, without any large pretrained language model, can meaningfully separate faithful from hallucinated LLM responses in Bangla, offering a

computationally inexpensive baseline for low-resource hallucination detection.

Keywords—Hallucination Detection, Bangla Natural Language Processing, Low-Resource Language, Text Classification, TF-IDF, Bag-of-Words, Support Vector Machine

I. INTRODUCTION

Large language models (LLMs) are now widely deployed for tasks such as question answering, summarization, and open-domain dialogue. Despite their fluency, LLMs are known to *hallucinate producing* content that is plausible in style but factually unsupported by, or contradictory to, a given source [1]. In question-answering settings, a hallucinated answer is especially dangerous because it is delivered with the same confident tone as a correct one, giving the user no surface-level signal that the information is unreliable.

Most existing hallucination-detection research has concentrated on high-resource languages, chiefly English. Benchmarks such as HaluEval [2] provide large collections of LLM-generated hallucinated and non-hallucinated samples, but they are not designed for Bangla, a language spoken natively by more than 230 million people. Bangla-specific evaluation of LLM hallucination has only very recently begun to receive attention: BenHalluEval [3] introduces the first

dedicated Bengali hallucination benchmark, covering generative question answering, code-mixed question answering, summarization, and reasoning, using an LLM-as-judge protocol over LLM-generated hallucinated candidates. While valuable as an evaluation framework, this line of work is oriented toward *assessing* how much a given LLM hallucinates rather than toward building a small, deployable classifier that can flag an individual answer as faithful or hallucinated against a supplied reference.

This distinction matters in practice. LLM-as-judge evaluation typically requires calling another large model at inference time, which is costly and often infeasible for resource-constrained applications. A lightweight classifier trained on classical lexical features, by contrast, can be trained and deployed on ordinary hardware, offering a practical first line of defense for Bangla question-answering systems, even if as the results in this paper show it cannot fully replace deeper semantic verification.

This paper makes the following contributions:

- We construct a labeled Bangla dataset of question, reference-answer, and LLM-answer triples, each annotated as Faithful or Hallucinated.
- We design a Bangla-specific text-normalization pipeline that explicitly handles Bangla punctuation marks not covered by standard punctuation regular expressions.
- We compare Bag-of-Words and TF-IDF representations, each built from unigram and bigram tokens of the concatenated reference and LLM answer, across three classical classifiers.
- We introduce a question-grouped train/test split that keeps every answer variant of a given question on the same side of the split, preventing the data leakage that a naive row-level split would introduce.
- We report Accuracy, Precision, Recall, and F1 on the Hallucinated class for all six (representation $\times$ classifier) configurations and analyze the Precision–Recall trade-off of each.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset. Section IV details the methodology. Section V reports experimental results. Section VI discusses the findings, Section VII outlines limitations and future work, and Section VIII concludes the paper.

II. RELATED WORK

A. Hallucination in Large Language Models

Huang *et al.* [1] provides a comprehensive survey of LLM hallucination, organizing causes into data-related, training-related, and inference-related factors, and surveying detection and mitigation strategies. Li *et al.* [2] introduce HaluEval, a large-scale benchmark of human and model-generated hallucinated samples for question answering, dialogue, and summarization, used to prove whether LLMs can recognize hallucinated content. Both works, however, are constructed for English.

B. Hallucination Detection for Bangla

BenHaluEval [3] is, to our knowledge, the first dedicated hallucination benchmark for Bengali. It constructs 12,000 hallucinated candidates using GPT-5.4 across twelve task-specific hallucination types drawn from three existing Bengali datasets and evaluates several LLMs under a dual-track protocol that separately measures false-positive rate on correct instances and hallucination-detection rate on hallucinated instances. This benchmark is designed to evaluate LLMs as judges of their own or other models' output, rather than to train a standalone lightweight classifier that verifies a candidate's answer against a specific reference passage in the gap this paper addresses.

C. Bangla Question-Answering Resources

BanglaRQA [4] is a benchmark reading-comprehension dataset containing 14,889 question–answer pairs over 3,000 Bangla Wikipedia passages, including both answerable and unanswerable questions with diverse answer types (single-span, multi-span, and yes/no). Although BanglaRQA was not constructed for hallucination labeling, its reference-grounded question–passage–answer structure and in particular its explicit inclusion of unanswerable questions reflects the same underlying idea this paper relies on that an LLM answer can only be judged faithful or hallucinated relative to an explicit reference passage.

D. Classical Text Representation and Classification

Bag-of-Words and TF-IDF weighting [5] remain widely used for lexical representations for text classification due to their simplicity and interpretability. Multinomial Naive Bayes [7], linear Support Vector Machines [6], and L1-regularized (LASSO) logistic regression [8], fitted efficiently via coordinate descent as implemented in glmnet [9], are standard baseline classifiers for such representations. In this paper, these classical methods, together with the quanteda text-mining toolkit for R [10], are applied to the Bangla hallucination-detection task as a computationally inexpensive alternative to LLM-based evaluation.

III. DATASET

A. Data Description

The dataset used in this study consists of 1,000 Bangla question answer records. Each record contains a question, a short reference passage, a gold reference answer, an answer generated by an LLM, and a binary label indicating whether the LLM answer is Faithful (supported by the reference passage) or Hallucinated (unsupported by, or contradictory to, the reference passage). The schema is summarized in Table I.

TABLE I
Dataset Schema

Field	Description
ID	Unique record identifier
Question	Bangla question text
Passage	Reference passage the question is grounded in
Ref. Answer	Gold answer supported by the passage
LLM Answer	Answer produced by an LLM for the question
Label	Faithful or Hallucinated

B. Class Distribution and Data Quality

The dataset was checked for missing values and exact duplicate rows prior to modeling. The class distribution of the Label column is Faithful (433) Hallucinated (567) and Faithful

(0.4330) Hallucinated (0.5670). As a label-quality check, an exact-match indicator between the reference and LLM answer was cross-tabulated against the true label; an exact match is expected to coincide with the faithful class in the great majority of cases, confirming the reliability of the annotation.

C. Answer-Pair Construction

For every record, the reference answer and the LLM answer are concatenated into a single text unit, *answer-pair text*, which is used as the unit of classification throughout this paper. Concatenation is necessary because hallucination is defined by the *relationship* between the two answers rather than by either text in isolation: vectorizing the LLM answer alone would discard the reference against which faithfulness is judged.

IV. METHODOLOGY

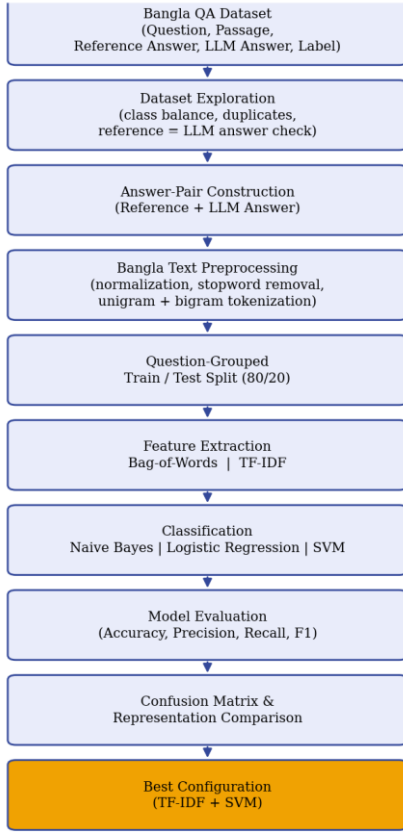


Fig. 1. End-to-end pipeline: from the labeled Bangla QA dataset to the best-performing (representation, classifier) configuration.

A. Text Preprocessing

Bangla text requires normalization beyond what standard, English-oriented regular-expression punctuation classes handle. The preprocessing pipeline removes control characters, standard punctuation, and Bangla-specific punctuation marks the দাঁড়ি (।), ডবল দাঁড়ি (॥), curly quotation marks, and en/em dashes none of which are matched by the generic punctuation character class. Consecutive whitespace produced by these substitutions is then collapsed to a single space. The cleaned text is tokenized, Bangla stopwords are removed using the *stopwords-iso* Bangla list,

and unigram and bigram tokens ($n = 1, 2$) are generated jointly, so that two-word sequences such as “জাতীয় ফুল” are retained as single features alongside their constituent unigrams.

B. Feature Representation

Two lexical representations are compared. Bag-of-Words (BoW) counts the occurrences of each unigram/bigram in each answer-pair text, forming a document–feature matrix. TF-IDF reweights the same counts by inverse document frequency,

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log_{10}(N / \text{df}(t)),$$

down-weighting terms that occur in many documents and up-weighting terms that are comparatively rare but discriminative. To control vocabulary size, terms occurring fewer than two times overall or in fewer than two documents are pruned from the training vocabulary, with a fallback to an unpruned vocabulary if fewer than 20 features would otherwise remain. The IDF weights and the vocabulary itself are computed from the training split only and then applied unchanged to the test split, so that no information from the test set influences feature construction.

C. Question-Grouped Train/Test Split

Because each question in the dataset is associated with more than one answer-pair row (typically one Faithful and one or more Hallucinated variants of the same question and passage), a row-level random split risks placing different rows of the *same* question on both sides of the split. A model could then partially memorize a specific question rather than learn generalizable lexical indicators of hallucination a form of data leakage. To prevent this, the split is performed at the *question level*: 20% of unique questions are randomly selected for testing, and every row belonging to a selected question is assigned to the test set, with all remaining rows forming the training set.

D. Classification Models

Three classical classifiers are trained on each feature representation. Multinomial Naive Bayes models $P(\text{word} | \text{class})$ under a word-independence assumption and predicts the class with the highest posterior probability. L1-regularized (LASSO) Logistic Regression is fitted via 10-fold cross-validated coordinate descent (glmnet, $\alpha = 1$), which shrinks uninformative feature weights to exactly zero, performing implicit feature selection; the regularization strength that minimizes cross-validated classification error ($\lambda_{\min}$) is used at test time. A linear-kernel Support Vector Machine (SVM) seeks the maximum-margin hyperplane separating the two classes in the (unscaled) BoW/TF-IDF feature space; features are left unscaled because their relative magnitudes already carry TF-IDF weighting information.

E. Evaluation Metrics

Predictions are scored against the Hallucinated class as the positive class. Given true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN):

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{FP} + \text{FN} + \text{TN}),$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}), \quad \text{Recall} = \text{TP} / (\text{TP} + \text{FN}),$$

$$\text{F1} = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}).$$

Because the dataset is class-imbalanced (Section III-B), F1 on the Hallucinated class rather than raw accuracy is treated as the primary metric: a trivial classifier that always predicts

Hallucinated would already attain an accuracy equal to the Hallucinated class proportion without learning anything.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

After the question-grouped split, the training and test sets contained training (807) test (193) rows respectively, with class distributions training labels Faithful (347) Hallucinated (460) and test labels Faithful (86) Hallucinated (107). The trimmed training vocabulary contains 616 unigram/bigram features.

B. Overall Classification Performance

Table II reports Accuracy, Precision, Recall, and F1 for all six (representation $\times$ classifier) configurations, sorted by F1 on the Hallucinated class. The best-performing configuration is TF-IDF + SVM, with F1 = 0.8016 and Accuracy = 0.7358.

TABLE II
Classification Performance (Hallucinated Class)

Rep.	Model	Acc.	Prec.	Rec.	F1
TF-IDF	SVM	0.7358	0.6867	0.9626	0.8016
BoW	SVM	0.6943	0.6538	0.9533	0.7757
BoW	LR	0.6736	0.6294	1.0000	0.7726
TF-IDF	LR	0.6736	0.6294	1.0000	0.7726
BoW	NB	0.6477	0.7600	0.5327	0.6264
TF-IDF	NB	0.6425	0.7500	0.5327	0.6230

LR = Logistic Regression; NB = Naive Bayes. Best configuration shaded.

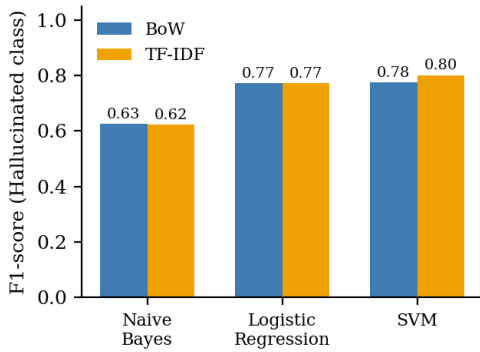


Fig. 2. F1-score (Hallucinated class) by representation and classifier.

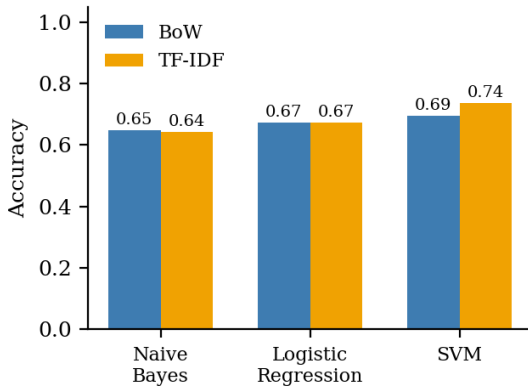


Fig. 3. Accuracy by representation and classifier.

C. Representation-Level Comparison

Averaging F1 and Accuracy across the three classifiers isolates the effect of the feature representation itself (Table III). TF-IDF attains a higher mean F1 (0.7324 vs. 0.7249) and a higher mean Accuracy (0.6840 vs. 0.6719) than BoW, and as Table II shows never underperforms BoW for the same classifier. The margin is modest, consistent with the answer-pair texts being short, which limits how much IDF re-weighting can differentiate common from rare terms.

TABLE III
Mean Performance by Representation

Representation	Mean F1	Mean Acc.
BoW	0.7249	0.6719
TF-IDF	0.7324	0.6840

D. Precision-Recall Trade-off

Fig. 4 plots Precision against Recall for all six configurations. The two SVM configurations and, more strongly, both Logistic Regression configurations sit close to the top-left region of high Recall, indicating that these models flag nearly all truly hallucinated answers but at the cost of also flagging some Faithful answers as hallucinated (lower Precision). Both Naive Bayes configurations sit in the opposite region: comparatively high Precision (0.75–0.76) but markedly lower Recall (0.5327), indicating a conservative classifier that misses roughly half of the true hallucinations while rarely raising a false alarm. Notably, BoW + Logistic Regression and TF-IDF + Logistic Regression produce numerically identical Precision, Recall, and F1 (Section VI), the only such exact tie among the six configurations.

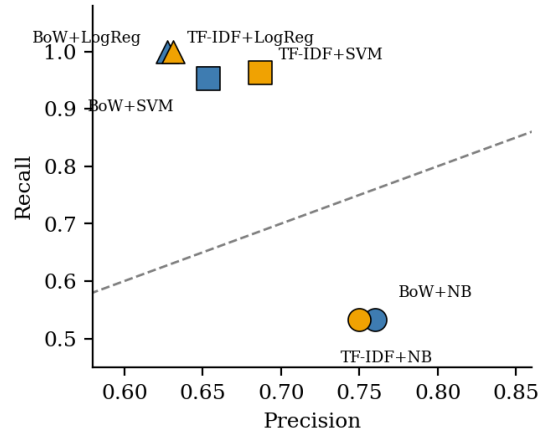


Fig. 4. Precision-Recall trade-off across all six (representation $\times$ classifier) configurations. The dashed line marks Precision = Recall.

E. Confusion Matrix of the Best Configuration

The confusion matrix of the best configuration (TF-IDF + SVM) is the confusion matrix of the best configuration (TF-IDF + SVM) is TP (103) FP (47) FN (4) TN (39), consistent with the high Recall and moderate Precision reported in Table II., consistent with the high Recall and moderate Precision reported in Table II.

VI. DISCUSSION

TF-IDF vs. BoW: TF-IDF's consistent but small advantage over BoW (Table III) is consistent with the down-weighting of ubiquitous Bangla function words and

boilerplate phrasing, while its magnitude is limited by the short length of the answer-pair texts, which caps how much a term's document frequency can vary across the corpus.

SVM's dominance: The linear SVM achieves the highest F1 under both representations, with the highest Recall of any classifier (0.9533–0.9626). This suggests that a linear boundary over lexical-overlap-style features most directly, the presence of tokens in the LLM answer that are absent from the reference answer captures a substantial share of the signal that distinguishes hallucinated from faithful answers, at the cost of a non-trivial false-positive rate on Faithful answers that happen to be phrased differently from the reference.

The Logistic Regression anomaly: BoW and TF-IDF + Logistic Regression produce exactly the same Precision, Recall, and F1 (Table II), with Recall = 1.0000. Because Recall = 1.0000 occurs only when a classifier predicts the positive class for every truly positive test instance and the identical Precision under two different feature weightings further suggests the fitted model relies on very few discriminative coefficients after L1 shrinkage the most likely explanation is that the cross-validated LASSO penalty ($\alpha = 1$) drove nearly all feature weights toward zero, leaving a near-intercept-only model that behaves similarly to a majority-class predictor regardless of which weighting scheme feeds it. This is a genuine limitation of the current configuration rather than evidence that Logistic Regression is representation-invariant, and is discussed further in Section VII.

Naive Bayes' conservatism: Naive Bayes attains the highest Precision (0.75–0.76) but the lowest Recall (0.5327) and F1 of the three classifiers. Its conditional-independence assumption is more strongly violated once bigrams are included alongside unigrams, since a bigram partially encodes the co-occurrence of its constituent unigrams; this likely contributes to its comparatively weaker overall performance despite its precision advantage.

Class imbalance: The dataset's class imbalance motivates the choice of F1 on the Hallucinated class as the primary metric rather than raw accuracy: a trivial always-Hallucinated predictor would already score close to the Hallucinated class's base rate on accuracy alone, which is precisely the behavior the Logistic Regression anomaly above appears to approximate.

VII. LIMITATIONS AND FUTURE WORK

- **Dataset scale:** with 1,000 question–answer pairs, the dataset is sufficient for classical machine learning but modest relative to the scale typically used to fine-tune transformer-based models, limiting vocabulary coverage.
- **Lexical-only representation:** BoW and TF-IDF on the concatenated answer pair cannot represent deeper semantic entailment; a faithful answer that paraphrases the reference with little lexical overlap, or a hallucinated answer that reuses reference vocabulary while altering a key fact such as a date or name, is difficult for either representation to separate correctly.
- **Regularization sensitivity:** the Logistic Regression anomaly discussed in Section VI suggests that the cross-validation criterion and the choice of $\lambda_{\min}$ merit further tuning, or a comparison against λ_{1se} , before

Logistic Regression can be considered a reliable baseline on this dataset.

- **General-purpose stopword list:** the Bangla stopword list used (stopwords-iso) is not tuned to this question-answering domain and may retain or remove tokens sub-optimally relative to a domain-specific list.
- **Single random split:** although the question-grouped split prevents leakage, a single 80/20 split gives one point estimate of performance; question-level k-fold cross-validation would provide a more stable estimate and confidence intervals.

Future work should incorporate explicit reference–answer overlap features (e.g., token Jaccard similarity, novel-token ratio, and TF-IDF cosine similarity between the reference and LLM answer) alongside the lexical vectors used here, since these features more directly operationalize the notion of “unsupported content” that defines hallucination. Incorporating contextual embeddings from a pretrained Bangla language model such as BanglaBERT as a stronger baseline and scaling the labeled set using a GPT-based hallucinated-answer generation protocol similar to that used in BenHalluEval [3], are also promising directions.

VIII. CONCLUSION

This paper presented a lightweight, classical machine-learning pipeline for detecting hallucination in Bangla LLM-generated answers, comparing Bag-of-Words and TF-IDF feature representations across Naive Bayes, Logistic Regression, and SVM classifiers under a leakage-safe, question-grouped train/test split. TF-IDF combined with a linear SVM achieved the best performance, with an F1-score of 0.8016 and an accuracy of 73.58% on the Hallucinated class, and TF-IDF consistently if modestly outperformed BoW across all three classifiers. These results demonstrate that simple lexical-overlap-based features, without recourse to a large pretrained language model or an LLM-as-judge pipeline, can meaningfully distinguish faithful from hallucinated answers in Bangla question answering, providing a computationally inexpensive baseline that future work can extend with semantic-overlap features and contextual embeddings.

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References

- [1] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu, “A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions,” *ACM Trans. Inf. Syst.*, vol. 43, no. 2, pp. 1–55, Jan. 2025.
- [2] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, and J.-R. Wen, “HaluEval: A large-scale hallucination evaluation benchmark for large language models,” in *Proc. 2023 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, Singapore, Dec. 2023, pp. 6449–6464.
- [3] S. E. S. Adib, A. A. Sani, E. A. Esham, A. Abrar, I. Tashdeed, and M. T. A. Chowdhury, “BenHalluEval: A multi-task hallucination evaluation framework for large language models on Bengali,” *arXiv:2605.31483*, 2026.
- [4] S. M. S. Ekram, A. A. Rahman, M. S. Altaf, M. S. Islam, M. M. Rahman, M. M. Rahman, M. A. Hossain, and A. R. M. Kamal, “BanglaRQA: A benchmark dataset for under-resourced Bangla language reading comprehension-based question answering with

diverse question-answer types,” in Findings of the Assoc. for Computational Linguistics: EMNLP 2022, Abu Dhabi, UAE, Dec. 2022, pp. 2518–2532.

- [5] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Inf. Process. Manage.*, vol. 24, no. 5, pp. 513–523, 1988.
- [6] C. Cortes and V. Vapnik, “Support-vector networks,” *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995.
- [7] A. McCallum and K. Nigam, “A comparison of event models for naive Bayes text classification,” in *AAAI-98 Workshop on Learning for Text Categorization*, 1998, pp. 41–48.
- [8] R. Tibshirani, “Regression shrinkage and selection via the lasso,” *J. R. Stat. Soc. Series B*, vol. 58, no. 1, pp. 267–288, 1996.
- [9] J. Friedman, T. Hastie, and R. Tibshirani, “Regularization paths for generalized linear models via coordinate descent,” *J. Stat. Softw.*, vol. 33, no. 1, pp. 1–22, 2010.
- [10] K. Benoit, K. Watanabe, H. Wang, P. Nulty, A. Obeng, S. Müller, and A. Matsuo, “quanteda: An R package for the quantitative analysis of textual data,” *J. Open Source Softw.*, vol. 3, no. 30, p. 774, 2018.